

Ashish Ranjan

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Final-year CSE (AI ML) student at VIT Bhopal University with experience in Java Full Stack Development, Machine Learning and Computer Vision. Passionate about building scalable backend systems and AI-powered applications using Spring Boot, React and modern ML frameworks. Seeking Software Engineer or Machine Learning Engineer opportunities.

EDUCATION

VIT Bhopal University

Bachelor of Technology – Computer Science & Engineering (AI & ML)

• **CGPA: 8.86 / 10.0**

Dr. Jagannath Mishra College

Intermediate (Class XII)

• **Percentage: 78.2%**

Disha Delphi Public School

Matriculation (Class X)

• **Percentage: 83.6%**

Sehore, MP

September 2023 – May 2027

Muzaffarpur, Bihar

April 2021 – March 2022

Kota, Rajasthan

April 2019 – March 2020

TECHNICAL SKILLS

Languages Java, Python, C++, JavaScript, SQL, C

Backend Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs, JWT, FastAPI, Flask, Maven

Frontend React.js, HTML, CSS, Tailwind CSS, shadcn/ui

Databases MySQL, SQL Server, SQLite, SQLAlchemy, JDBC, JPA/Hibernate

Machine Learning & AI Scikit-learn, Pandas, NumPy, Matplotlib, YOLO11, DeepFace, FaceNet512, TensorFlow, PyTorch, OpenCV, Random Forest, SVM

Developer Tools Git, GitHub, IntelliJ IDEA, VS Code, Swagger/OpenAPI, Postman

Coursework Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Object-Oriented Programming, Cloud Computing, Software Engineering

Core CS Arrays, Strings, Linked Lists, Trees, Graphs, Hashing, Sorting & Searching

EXPERIENCE

MPOnline Limited

Software Development & AI/ML Trainee

May 2026 – July 2026

Bhopal, MP

- Developed a web-based Library Management System using ASP.NET Core MVC, C#, Entity Framework Core and SQL Server following the MVC architecture.
- Implemented role-based authentication, book management, borrowing workflows and CRUD operations with optimized relational database design.
- Built multiple machine learning workflows involving data preprocessing, feature engineering, model training and evaluation using Scikit-learn, Pandas and NumPy.
- Applied SQL and PL/SQL concepts including joins, stored procedures, functions and triggers for efficient database operations.

PROJECTS

SmartRetailAI | [GitHub](#)

Python, FastAPI, YOLO11, DeepFace, FaceNet512, Scikit-learn, SQLAlchemy

- * Developed an AI-powered retail platform integrating face recognition and **YOLO11**-based product detection through **FastAPI** services.
- * Implemented customer identification using **FaceNet512** embeddings, **DeepFace** feature extraction and an **SVM classifier**.
- * Built end-to-end customer visit workflows with **SQLAlchemy** and **REST APIs** for customer tracking and review management.
- * Designed the system to support scalable AI modules including sentiment analysis, **chatbot** and **personalized recommendations**.

Hospital Management System | [GitHub](#) | *Java, Spring Boot, Spring Security, JWT, React, MySQL*

- * Built a full-stack Hospital Management System with **JWT** authentication and **role-based access** for Admin, Doctor, Receptionist and Patient.
- * Developed REST APIs for appointments, patients, billing and medical records using **Spring Boot, JPA and MySQL**.
- * Implemented appointment scheduling, rescheduling, cancellation and doctor availability management.
- * Designed layered architecture with Swagger documentation, validation and centralized exception handling.

AI Powered Sentiment Analyzer | [GitHub](#) | *Java, Spring Boot, HuggingFace API, MySQL, JUnit 5*

- * Integrated **HuggingFace Inference API** to classify text into Positive, Neutral and Negative sentiments with confidence scores.
- * Built REST APIs for inference, JSON parsing and result storage using Spring Boot and MySQL.
- * Processed 500+ Yelp reviews through resilient batch pipelines with retry and rate-limiting strategies.
- * Implemented **JUnit 5 tests** with mocked API responses to validate business logic.

Real-Time Credit Card Fraud Detection System | [GitHub](#) | *Scikit-learn, SVM, Python*

- * Built a real-time fraud detection classifier for high-dimensional, imbalanced transaction data using **SVMs** (linear, RBF, polynomial kernels), validated with stratified *k*-fold cross-validation
- * Tuned hyperparameter via GridSearchCV to optimize and achieve **0.91 precision-recall AUC value** on a rare-event classification problem
- * Reduced dimensionality and training time via recursive feature elimination and correlation-based feature selection
- * Delivered a production deployment recommendation based on kernel and cost-tradeoff comparison

CERTIFICATIONS

Machine Learning with Python – University of Michigan	Dec 2025
Skills: Scikit-Learn, Regression, Classification, Clustering, Model Evaluation	
NPTEL – Cloud Computing (Elite Grade)	Jun 2025
Cloud architectures, deployment models, scalable system design	
NPTEL – Internet of Things (Overall Score: 90)	Jun 2026
IoT architecture, embedded systems, IoT communication protocols	
Deloitte Australia – Data Analytics Job Simulation	Jul 2025
Tableau Dashboard, Excel data classification, business insights	

AREAS OF INTEREST

Java Full Stack Development		Cloud Engineering		Machine Learning		Data Structures and Algorithms
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